

Solve all of these limits :

حل كلاً من النهايات الآتية :

1) $\lim_{x \rightarrow 0} \frac{\sin 2x}{\sqrt{x+1} - 1}$

2) $\lim_{x \rightarrow 0} \frac{\sin 2x}{\sqrt{2x+3} - \sqrt{3}}$

3) $\lim_{x \rightarrow 0} \frac{\sin 3x}{\sqrt{x+2} - \sqrt{2}}$

4) $\lim_{x \rightarrow 0} \frac{\sqrt{1+\sin 2x} - \sqrt{1-\sin 2x}}{x}$

5) $\lim_{x \rightarrow 0} \frac{\sqrt{1+x^2} - \cos x}{x^2}$

6) $\lim_{x \rightarrow 0} \frac{\sqrt{1+\sin^2 x} - \cos x}{\sin^2 x}$

7) $\lim_{x \rightarrow 0} \frac{\sin^2 x}{\sqrt{1+x \sin x} - \cos x}$

8) $\lim_{x \rightarrow 0} \frac{\sqrt{x+9} - 3}{\sin 7x}$

9) $\lim_{x \rightarrow 0} \frac{\sqrt{1+\sin x} - \sqrt{1-\sin x}}{\tan x}$

10) $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$

11) $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$

12) $\lim_{x \rightarrow 0} \frac{1 - \cos 3x}{x^2}$

13) $\lim_{x \rightarrow 0} \frac{1 - \cos 4x}{x^2}$

14) $\lim_{x \rightarrow 0} \frac{1 - \cos ax}{x^2}$

15) $\lim_{x \rightarrow 0} \frac{x \sin 3x}{1 - \cos x}$

16) $\lim_{x \rightarrow 0} \frac{1 - \cos x + \sin x}{x}$

17) $\lim_{x \rightarrow 0} \frac{1 - \cos 3x}{1 - \cos 5x}$

18) $\lim_{x \rightarrow 0} \frac{1 + \sin x - \cos x}{1 - \sin x - \cos x}$

19) $\lim_{x \rightarrow 0} \frac{\sqrt{2} - \sqrt{1 - \cos x}}{\tan^2 x}$

20) $\lim_{x \rightarrow 0} \frac{\sqrt{1+\tan x} - \sqrt{1-\tan x}}{\sin 2x}$

21) $\lim_{x \rightarrow 0} \frac{2x - \sin x}{\sqrt{1 - \cos x}}$

22) $\lim_{x \rightarrow 0} \frac{1 - \sqrt{\cos x}}{\tan^2 x}$

23) $\lim_{x \rightarrow 0} \frac{1 + x - \cos 2x}{1 - \sqrt{1 + \sin x}}$

24) $\lim_{x \rightarrow 0} \frac{(a-b)x}{\sin ax - \sin bx}$

25) $\lim_{x \rightarrow 0} \frac{1 - \cos x \cos 2x}{x^2}$

26) $\lim_{x \rightarrow 0} \frac{\cos^2 x - \cos x}{x^2}$

27) $\lim_{x \rightarrow 0} \frac{1 - \cos x \sqrt{\cos 2x}}{x^2}$

28) $\lim_{x \rightarrow 0} \frac{x \sin 2x}{1 - \cos x}$

29) $\lim_{x \rightarrow 0} \frac{\cos x - \sqrt{\cos 2x}}{\sin^2 x}$

30) $\lim_{x \rightarrow 0} \frac{x}{x + \sin x}$

31) $\lim_{x \rightarrow 0} \frac{\sin x}{\sqrt{x}}$

32) $\lim_{x \rightarrow 0} \frac{\sin^2 9x - x^2}{1 - \cos 2x}$

33) $\lim_{x \rightarrow 0} \frac{1 - \cos x}{\sin x}$

34) $\lim_{x \rightarrow -\frac{\pi}{4}} \frac{\sin x + \cos x}{x + \frac{\pi}{4}}$

35) $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x}{\sqrt[3]{1 - \sin x}}$

36) $\lim_{x \rightarrow 0} \frac{x^2 - 1 + \cos^2 x}{x \sin x}$

37) $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin x - 1}{x - \frac{\pi}{2}}$

انتهت الأسئلة.

$$\lim_{x \rightarrow 0} \left(\frac{2 \sin^2 \frac{x}{2}}{x} + \frac{\sin x}{x} \right) = \lim_{x \rightarrow 0} \left(\frac{\sin \frac{x}{2}}{\frac{x}{2}} \sin \frac{x}{2} + \frac{\sin x}{x} \right)$$

$$= (1)(0) + 1 = 1$$

$$17) \lim_{x \rightarrow 0} \frac{1 - \cos 3x}{1 - \cos 5x} \left(\frac{1 + \cos 3x}{1 + \cos 5x} \right) \left(\frac{1 + \cos 5x}{1 + \cos 3x} \right)$$

$$= \lim_{x \rightarrow 0} \frac{1 - \cos^2 3x}{1 - \cos^2 5x} \left(\frac{1 + \cos 5x}{1 + \cos 3x} \right)$$

$$= \lim_{x \rightarrow 0} \frac{\sin^2 3x}{\sin^2 5x} \left(\frac{1 + \cos 5x}{1 + \cos 3x} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{\sin 3x}{\sin 5x} \right)^2 \left(\frac{1 + \cos 5x}{1 + \cos 3x} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{3 \frac{\sin 3x}{3x}}{5 \frac{\sin 5x}{5x}} \right)^2 \left(\frac{1 + \cos 5x}{1 + \cos 3x} \right) = \left(\frac{3}{5} (1) \right)^2 \frac{2}{2} = \frac{9}{25}$$

$$18) \lim_{x \rightarrow 0} \frac{1 + \sin x - \cos x}{1 - \sin x - \cos x}$$

$$1 - \cos x = 2 \sin^2 \frac{x}{2}$$

$$\lim_{x \rightarrow 0} \frac{1 + \sin x - \cos x}{1 - \sin x - \cos x} = \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2} + \sin x}{2 \sin^2 \frac{x}{2} - \sin x}$$

نقسم البسط والمقام على x :

$$\lim_{x \rightarrow 0} \frac{2 \frac{\sin^2 \frac{x}{2}}{x} + \frac{\sin x}{x}}{2 \frac{\sin^2 \frac{x}{2}}{x} - \frac{\sin x}{x}}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{\sin \frac{x}{2}}{\frac{x}{2}} \sin \frac{x}{2} + \frac{\sin x}{x}}{\frac{\sin \frac{x}{2}}{\frac{x}{2}} \sin \frac{x}{2} - \frac{\sin x}{x}} = \frac{(1)(0) + 1}{(1)(0) - 1} = -1$$

$$19) \lim_{x \rightarrow 0} \frac{\sqrt{2} - \sqrt{1 + \cos x}}{\tan^2 x}$$

$$= \lim_{x \rightarrow 0} \frac{2 - 1 - \cos x}{\tan^2 x} \cdot \frac{1}{\sqrt{2} + \sqrt{1 + \cos x}}$$

$$= \lim_{x \rightarrow 0} \frac{1 - \cos x}{\tan^2 x} \cdot \frac{1}{\sqrt{2} + \sqrt{1 + \cos x}}$$

$$11) \lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2} = \lim_{x \rightarrow 0} \frac{2 \sin^2 x}{x^2} = 2 \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^2 = 2$$

$$12) \lim_{x \rightarrow 0} \frac{1 - \cos 3x}{x^2} = \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{3x}{2}}{x^2}$$

$$= \lim_{x \rightarrow 0} 2 \left(\frac{\sin \frac{3x}{2}}{\frac{x}{2}} \right)^2 = \lim_{x \rightarrow 0} 2 \left(\frac{3}{2} \cdot \frac{\sin \frac{3x}{2}}{\frac{3x}{2}} \right)^2$$

$$= 2 \left(\frac{3}{2} \cdot 1 \right)^2 = \frac{9}{2}$$

$$13) \lim_{x \rightarrow 0} \frac{1 - \cos 4x}{x^2} = \lim_{x \rightarrow 0} \frac{2 \sin^2 2x}{x^2}$$

$$\sin 2x = 2 \sin x \cos x$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos 4x}{x^2} = \lim_{x \rightarrow 0} \frac{2(2 \sin x \cos x)^2}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{8 \sin^2 x \cos^2 x}{x^2} = \lim_{x \rightarrow 0} 8 \left(\frac{\sin x}{x} \right)^2 \cos^2 x$$

$$= (8)(1)(1) = 8$$

$$14) \lim_{x \rightarrow 0} \frac{1 - \cos ax}{x^2} = \lim_{x \rightarrow 0} \frac{1 - \cos ax}{x^2} \left(\frac{1 + \cos ax}{1 + \cos ax} \right)$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos^2 ax}{x^2(1 + \cos ax)} = \lim_{x \rightarrow 0} \frac{\sin^2 ax}{x^2(1 + \cos ax)}$$

$$\lim_{x \rightarrow 0} \left(a \frac{\sin ax}{ax} \right)^2 \frac{1}{1 + \cos x} = \frac{a^2}{2}$$

$$15) \lim_{x \rightarrow 0} \frac{x \sin 3x}{1 - \cos x} = \lim_{x \rightarrow 0} \frac{x \sin 3x}{1 - \cos x} \left(\frac{1 + \cos x}{1 + \cos x} \right)$$

$$= \lim_{x \rightarrow 0} \frac{x \sin 3x}{1 - \cos^2 x} (1 + \cos x)$$

$$= \lim_{x \rightarrow 0} \frac{x \sin 3x}{\sin^2 x} (1 + \cos x)$$

$$= \lim_{x \rightarrow 0} \frac{x}{\sin x} \frac{\sin 3x}{\sin x} (1 + \cos x)$$

$$= \lim_{x \rightarrow 0} \frac{x}{\sin x} \frac{3 \frac{\sin 3x}{3x}}{\frac{\sin x}{x}} (1 + \cos x) = (1) \frac{3(1)}{1} (2) = 6$$

$$16) \lim_{x \rightarrow 0} \frac{1 - \cos x + \sin x}{x} = \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2} + \sin x}{x}$$

$$\frac{\frac{x}{xq \sin \theta} - \frac{x}{xv \sin \theta}}{q - v} \lim_{\theta \rightarrow x} = \frac{xq \sin \theta - xv \sin \theta}{x(q - v)} \lim_{\theta \rightarrow x}$$

$$(24) \quad \frac{(a-b)x}{\lim_{x \rightarrow 0} \sin ax - \sin bx}$$

$$(23) \quad \lim_{x \rightarrow 0} \frac{1 + x - \cos 2x}{1 - \sqrt{1 + \sin x}} = \lim_{x \rightarrow 0} \frac{2 \sin^2 x + x}{\frac{1 + \sqrt{1 + \sin x}}{1 + \sqrt{1 + \sin x}}} =$$

$$\lim_{x \rightarrow 0} \frac{\tan^2 x}{1 - \sqrt{\cos x}} = \lim_{x \rightarrow 0} \frac{\tan^2 x}{1 + \sqrt{\cos x}} \cdot \frac{1 + \sqrt{\cos x}}{1 + \sqrt{\cos x}} \quad (22)$$

$$\lim_{x \rightarrow 0} \frac{z^2 - 1}{z^2 - 2z + 1} = \lim_{x \rightarrow 0} \frac{(z-1)(z+1)}{(z-1)^2} = \lim_{x \rightarrow 0} \frac{z+1}{z-1} = \frac{1}{-1} = -1$$

$$|x|_0 = 0 \leftarrow x \text{ is } 0 \text{ uis} - = |x \text{ uis}|$$

$$\lim_{x \rightarrow 0^+} \frac{x \sin x}{2x - \sin x} = \lim_{x \rightarrow 0^+} \frac{\sqrt{1 - \cos x}}{2x - \sin x} = \lim_{x \rightarrow 0^+} \frac{1}{2x - \sin x} \frac{x \sin x}{\sqrt{1 + \cos x}}$$

$$|\sin x| + 0 \leq x \leq |\sin x| \Rightarrow |\sin x| = x$$

$$\lim_{x \rightarrow 0} \frac{2x - \sin x}{x \cos x - 1} = \lim_{x \rightarrow 0} \frac{2x - \sin x}{x \cos x - 1} \cdot \frac{x \cos x + 1}{x \cos x + 1} \quad (21)$$

$$(20) \quad \lim_{x \rightarrow 0} \frac{\sin 2x}{\sqrt{1 + \tan x} - \sqrt{1 - \tan x}}$$

$$= \lim_{x \rightarrow 0} \frac{x \tan^2 x}{1 - \cos x} \cdot \frac{1 + \cos x}{1 + \cos x} \cdot \frac{\sqrt{1 + \cos x}}{\sqrt{1 + \cos x}} =$$

$$= \lim_{x \rightarrow 0} \frac{1 - \cos^2 x (2 \cos^2 x - 1)}{x^2} \cdot \frac{1}{1 + \cos x \sqrt{\cos 2x}}$$

$$= \lim_{x \rightarrow 0} \frac{1 - 2 \cos^4 x + \cos^2 x}{x^2} \cdot \frac{1}{1 + \cos x \sqrt{\cos 2x}}$$

ليكن $\cos x = t$

$$1 + t^2 - 2t^4 = 0$$

$$t_1 = 1, \quad t_2 = -1$$

بإجراء القسمة الإقليدية نجد:

$$1 + t^2 - 2t^4 = (1 - t)(1 + t)(2t^2 + 1)$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x \sqrt{\cos 2x}}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{(1 - \cos x)(1 + \cos x)(2 \cos^2 x + 1)}{x^2(1 + \cos x \sqrt{\cos 2x})}$$

$$= \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2} (1 + \cos x)(2 \cos^2 x + 1)}{x^2(1 + \cos x \sqrt{\cos 2x})}$$

$$= \lim_{x \rightarrow 0} 2 \left(\frac{\sin \frac{x}{2}}{2 \frac{x}{2}} \right)^2 \frac{(1 + \cos x)(2 \cos^2 x + 1)}{(1 + \cos x \sqrt{\cos 2x})}$$

$$= (2) \left(\frac{1}{2} (1) \right)^2 \frac{6}{2} = \frac{3}{2}$$

$$28) \lim_{x \rightarrow 0} \frac{x \sin 2x}{1 - \cos x} = \lim_{x \rightarrow 0} \frac{2x \sin x \cos x}{1 - \cos x} \cdot \frac{1 + \cos x}{1 + \cos x}$$

$$= \lim_{x \rightarrow 0} \frac{2x \sin x \cos x}{1 - \cos^2 x} \cdot \frac{1 + \cos x}{1}$$

$$= \lim_{x \rightarrow 0} \frac{2x \sin x \cos x}{\sin^2 x} \cdot \frac{1 + \cos x}{1}$$

$$= \lim_{x \rightarrow 0} \frac{2x \cos x}{\sin x} \cdot \frac{1 + \cos x}{1} = (2)(1)(1)(2) = 4$$

$$29) \lim_{x \rightarrow 0} \frac{\cos x - \sqrt{\cos 2x}}{\sin^2 x}$$

$$= \lim_{x \rightarrow 0} \frac{\cos^2 x - \cos 2x}{\sin^2 x (\cos x + \sqrt{\cos 2x})}$$

$$= \lim_{x \rightarrow 0} \frac{\cos^2 x - (\cos^2 x - \sin^2 x)}{\sin^2 x (\cos x + \sqrt{\cos 2x})}$$

$$= \lim_{x \rightarrow 0} \frac{\sin^2 x}{\sin^2 x (\cos x + \sqrt{\cos 2x})}$$

$$= \lim_{x \rightarrow 0} \frac{1}{(\cos x + \sqrt{\cos 2x})} = \frac{1}{2}$$

$$25) \lim_{x \rightarrow 0} \frac{1 - \cos x \cos 2x}{x^2}$$

$$\cos 2x = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x \cos 2x}{x^2} = \lim_{x \rightarrow 0} \frac{1 - \cos x (2 \cos^2 x - 1)}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{1 + \cos x - 2 \cos^3 x}{x^2}$$

ليكن $\cos x = t$

$$1 + t - 2t^3 = 0$$

$$t_0 = 1$$

بإجراء القسمة الإقليدية نجد:

$$1 + t - 2t^3 = (1 - t)(2t^2 + 2t + 1)$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x \cos 2x}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{(1 - \cos x)(2 \cos^2 x + 2 \cos x + 1)}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{(2 \sin^2 \frac{x}{2})(2 \cos^2 x + 2 \cos x + 1)}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{(2 \sin^2 \frac{x}{2})(2 \cos^2 x + 2 \cos x + 1)}{x^2}$$

$$= \lim_{x \rightarrow 0} 2 \left(\frac{\sin \frac{x}{2}}{x} \right)^2 (2 \cos^2 x + 2 \cos x + 1)$$

$$= \lim_{x \rightarrow 0} 2 \left(\frac{\sin \frac{x}{2}}{2 \frac{x}{2}} \right)^2 (2 \cos^2 x + 2 \cos x + 1)$$

$$= (2) \left(\frac{1}{2} (1) \right)^2 (5) = \frac{5}{2}$$

$$26) \lim_{x \rightarrow 0} \frac{\cos^2 x - \cos x}{x^2} = \lim_{x \rightarrow 0} \frac{1 - \cos x - \sin^2 x}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2} - \sin^2 x}{x^2} = \lim_{x \rightarrow 0} 2 \left(\frac{\sin \frac{x}{2}}{2 \frac{x}{2}} \right)^2 - \left(\frac{\sin x}{x} \right)^2$$

$$= \frac{1}{2} - 1 = -\frac{1}{2}$$

$$27) \lim_{x \rightarrow 0} \frac{1 - \cos x \sqrt{\cos 2x}}{x^2} \cdot \frac{1 + \cos x \sqrt{\cos 2x}}{1 + \cos x \sqrt{\cos 2x}}$$

$$= \lim_{x \rightarrow 0} \frac{1 - \cos^2 x \cos 2x}{x^2} \cdot \frac{1}{1 + \cos x \sqrt{\cos 2x}}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \sqrt[3]{(1 + \sin x) \cos x} = \sqrt[3]{2 \times 0} = 0$$

$$36) \lim_{x \rightarrow 0} \frac{x^2 - 1 + \cos^2 x}{x \sin x} = \lim_{x \rightarrow 0} \frac{x^2 - \sin^2 x}{x \sin x}$$

$$= \lim_{x \rightarrow 0} \frac{1 - \left(\frac{\sin x}{x}\right)^2}{\frac{\sin x}{x}} = \frac{1 - 1}{1} = 0$$

$$37) \lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin x - 1}{x - \frac{\pi}{2}}$$

ليكن التابع f المعرفة على $\mathbb{R}$ وفق :

$$f(x) = \sin x$$

$$f\left(\frac{\pi}{2}\right) = \sin \frac{\pi}{2} = 1$$

$$f'(x) = \cos x$$

$$f'\left(\frac{\pi}{2}\right) = \cos \frac{\pi}{2} = 0$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin x - 1}{x - \frac{\pi}{2}} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{f(x) - f\left(\frac{\pi}{2}\right)}{x - \frac{\pi}{2}} = f'\left(\frac{\pi}{2}\right) = 0$$

نعمب تعريف العدد المثنق .



$$30) \lim_{x \rightarrow 0} \frac{x}{x + \sin x} = \lim_{x \rightarrow 0} \frac{1}{1 + \frac{\sin x}{x}} = \frac{1}{2}$$

$$31) \lim_{x \rightarrow 0} \frac{\sin x}{\sqrt{x}} = \lim_{x \rightarrow 0} \frac{\sin x}{x} \sqrt{x} = (1)(0) = 0$$

$$32) \lim_{x \rightarrow 0} \frac{\sin^2 9x - x^2}{1 - \cos 2x} = \lim_{x \rightarrow 0} \frac{\sin^2 9x - x^2}{2 \sin^2 x}$$

نقسم البسط والمقام على x^2 :

$$\lim_{x \rightarrow 0} \frac{\sin^2 9x - x^2}{1 - \cos 2x} = \lim_{x \rightarrow 0} \frac{\frac{\sin^2 9x}{x^2} - 1}{2 \frac{\sin^2 x}{x^2}}$$

$$= \lim_{x \rightarrow 0} \frac{\left(9 \frac{\sin 9x}{9x}\right)^2 - 1}{2 \left(\frac{\sin x}{x}\right)^2} = \frac{81(1) - 1}{2(1)} = 40$$

$$33) \lim_{x \rightarrow 0} \frac{1 - \cos x}{\sin x} = \lim_{x \rightarrow 0} \frac{\sin^2 x}{\sin x (1 + \cos x)}$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{(1 + \cos x)} = \frac{0}{2} = 0$$

$$34) \lim_{x \rightarrow -\frac{\pi}{4}} \frac{\sin x + \cos x}{x + \frac{\pi}{4}}$$

$$\sin(a + b) = \sin a \cos b + \cos a \sin b$$

$$\lim_{x \rightarrow -\frac{\pi}{4}} \frac{\sin x + \cos x}{x + \frac{\pi}{4}} = \lim_{x \rightarrow -\frac{\pi}{4}} \sqrt{2} \frac{\frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{2}} \cos x}{x + \frac{\pi}{4}}$$

$$= \lim_{x \rightarrow -\frac{\pi}{4}} \sqrt{2} \frac{\cos \frac{\pi}{4} \sin x + \sin \frac{\pi}{4} \cos x}{x + \frac{\pi}{4}}$$

$$= \lim_{x \rightarrow -\frac{\pi}{4}} \sqrt{2} \frac{\sin(x + \frac{\pi}{4})}{x + \frac{\pi}{4}} = \sqrt{2}(1) = \sqrt{2}$$

$$35) \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x}{\sqrt[3]{(1 - \sin x) \frac{1 + \sin x}{1 + \sin x}}}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x}{\sqrt[3]{(1 - \sin^2 x) \frac{1}{1 + \sin x}}} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{\sqrt[3]{1 + \sin x} \cos x}{\sqrt[3]{\cos^2 x}}$$